



△ Emerald-cut diopside gem

Diopside

Diopside is one of the minerals found alongside diamonds in some kimberlite, an igneous rock, generally occurring as dark bottle-green, light green, brown, blue, or colourless stones. Rich green diopside, coloured by chromium and known as chrome diopside, is faceted as a prized collector's gem, and would potentially rival emerald in popularity but for its softness. Stones of another rich hue, a violet-blue coloured by manganese, are found in Italy and the USA – these are sometimes called violane and are highly prized collector's gems.

Specification

Chemical name Calcium, magnesium silicate | **Formula** $\text{CaMg}(\text{Si}_2\text{O}_6)$ | **Colours** White, pale to dark green, violet-blue
Structure Monoclinic | **Hardness** 5–6 | **SG** 3.2–3.4
RI 1.66–1.72 | **Lustre** Vitreous | **Streak** White to pale green
Locations Italy, USA, Myanmar, Austria, Canada, Pakistan, Sri Lanka

Diopside crystals



Diopside crystals | Rough | This specimen consists of a number of prismatic, green crystals of diopside embedded in a groundmass of quartz.



Violane | Rough | Rich purple diopside is sometimes called violane, and is shown here in massive form suitable for cutting into cabochons.



Chrome diopside | Colour variety | This deep green chrome diopside gemstone features a particularly fine colour, emphasized by its clean emerald cut.

Fibrous diopside material can be cut into cabochons to show cat's-eye patterning



Interesting flaws | Cut | Although this long, rectangular step-cut diopside features a number of flaws, they add to the character and interest of the gemstone.

Modified girdle facets



Modified brilliant | Cut | Viewed side-on, this richly coloured diopside can be seen to have modified girdle facets on the crown, and main facets on the pavilion.

White gold setting



Ring | Set | Dark green diopside is considered the most desirable colour, but lighter green, as in these matched stones set in a ring, has its own beauty.